Introduction

This test program intends to demonstrate the seal performance under site conditions after several reports of failure and inconsistency of leakage.

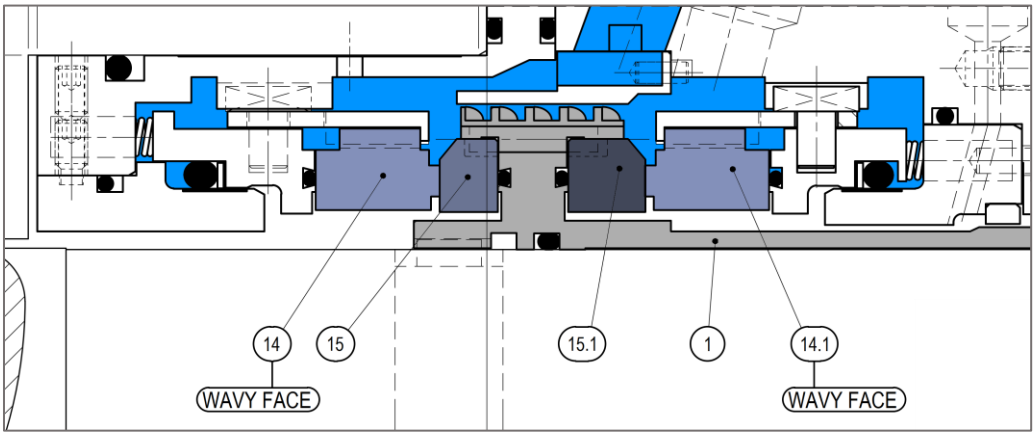
The subject of this testing is mechanical seal order 950969 modified. This version of mechanical seal was designed based on new seal faces geometry to include lasered wavy face technology.

Test N°2 – (Sep 01 – Sep 07, 2023)

Second test is a duplicate of test N001, this test will be performed with same faces from test 01 without lapping or re-conditioning.
Barrier fluid will be simulated as per site conditions MonoEthylene Glycol + Water (30%/70%).

Seal Configuration

Seal faces			
	Part Drawing	Material	Face condition
Sleeve (1)	C0619166NL	Alloy C-276	From Test 01
Inner Seal Stator (14)	C0618722SL	Silicon Carbide (SL)	From Test 01
Inner Seal Rotor (15)	C0619339SL	Silicon Carbide (SL)	From Test 01
Outer Seal Stator (14.1)	C0618722SL	Silicon Carbide (SL)	From Test 01
Outer Seal Rotor (15.1)	C0618723SL	Silicon Carbide (SL)	From Test 01



Cross Section of The New Parts

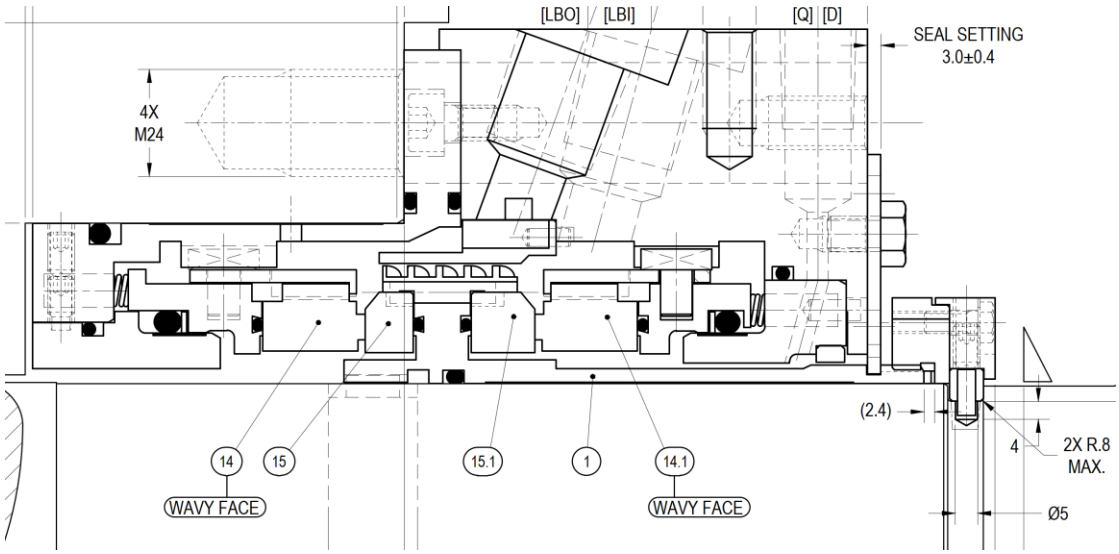
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Test Conditions & Results

	Job seal barrier
Duration	Minimum 144
Speed	2950 rpm
Fluid	Glycol 30/ Water 70
Pressure	70 barg
Inlet temp.	40 °C
Outlet temp.	43 °C
Inner Seal Leakage	Avg. 16.5 ml/h
Outer Seal Leakage	Avg. 17 ml/h
Flow	13.2-12.2 l/min

Tester Drawing D0571813



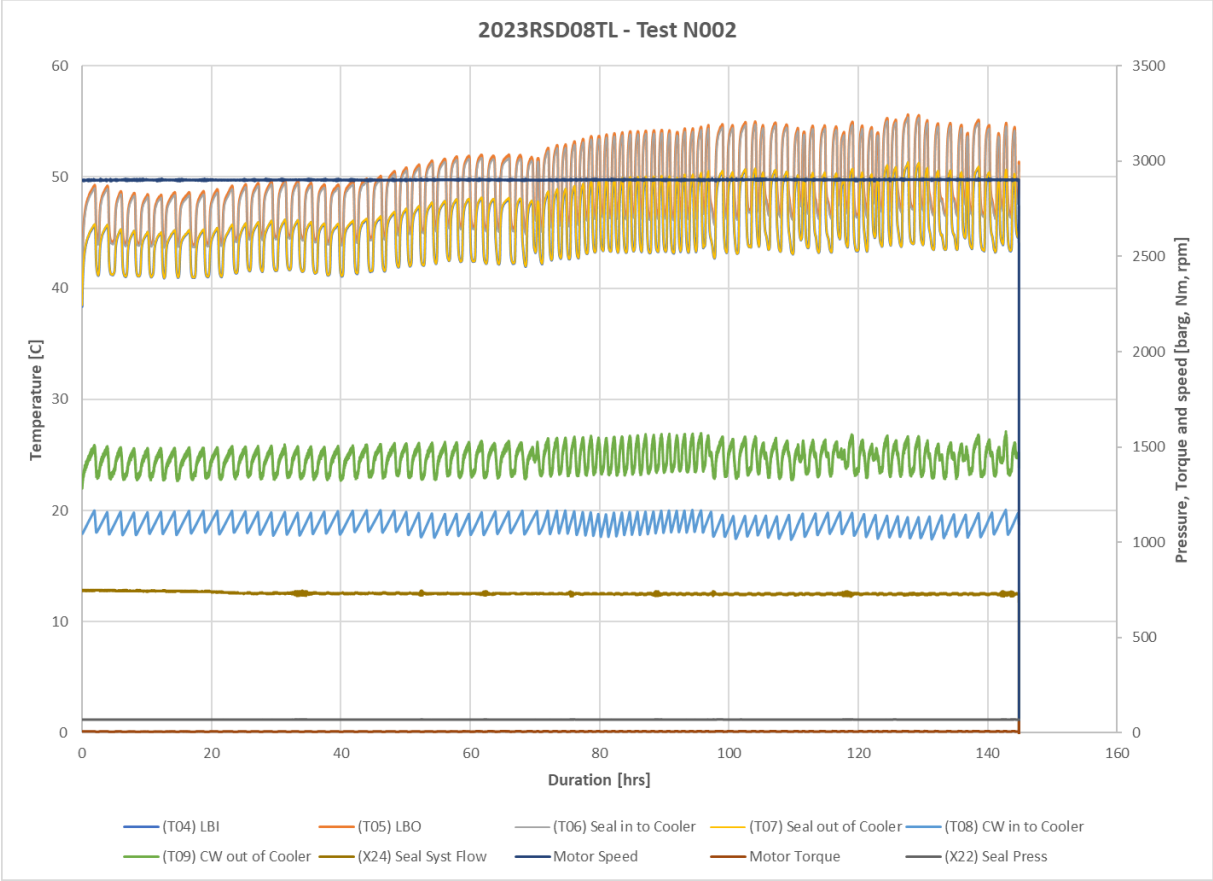
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Tester unit B

Logged Data

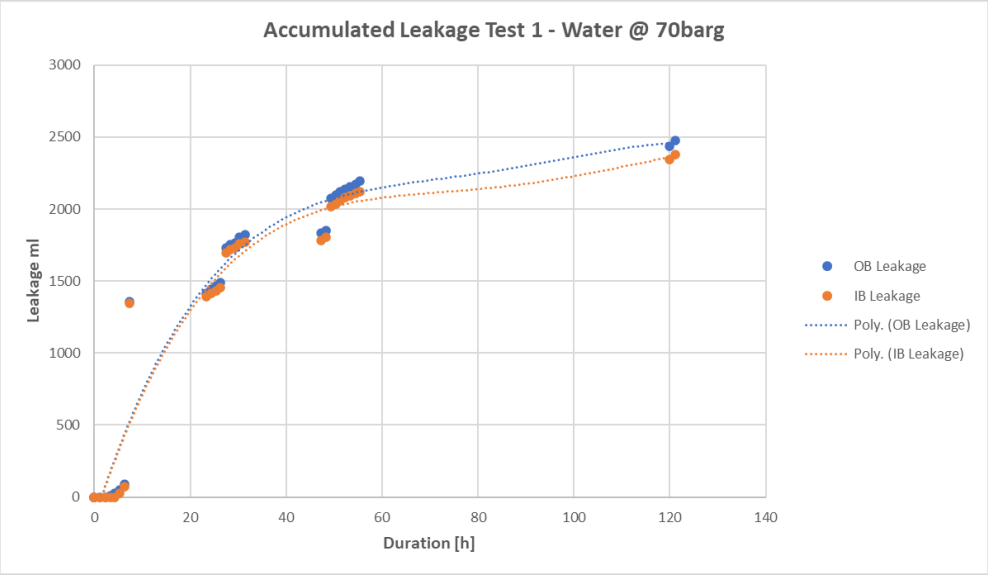


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The first 24h showed stable performance, however, barrier temperature was too low reason for reducing the cooling water flow to have higher barrier temperature.

Leakage was steady during the whole testing period.



Manually Logged Data

Test Project Number:		2023RSD08TL		Run no.:		TCN002					OB				IB				
Test Unit:		B		ire Rate s/S on 60 trigger on 5							Seal Type:		UHTW/UHTW		Seal Type:		UHTW/UHTW		
Speed:		CCW		Aligned		yes					Stationary Face:		C618722SL used		Stationary Face:		C618722SL used		
Pres / Temp:				Cooling CTRL Valve 0% is complete open							Rotating Face:		C0619323SL used		Rotating Face:		used		
Barrier Medium:		70%Demi water- 30%Glycol		Flow CTRL Valve 100% is complete open															
											OB				IB				
Date	Time	Running Hours Total	Runned Hours after a start	By	RPM	Torque [Nm]	Syst Flow [l/min]	Seal Press [barg]	LBI Temp. [°C]	LBO Temp. [°C]	OB total leakage in the glasss	OB Leakage [ml]	Over [hours]	OB leakage Rate [ml/hr]	IB total leakage in the glass	IB Leakage [ml]	Over [hours]	IB Leakage Rate [ml/hr]	
Friday, September 1, 2023	9:55	0.00	0.00	Rob	0	0.00	0.00	0.00	21.90	21.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
Friday, September 1, 2023	10:12	0.00	0.00	Rob		0.00	0.00	70.00	21.74	20.80	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
Friday, September 1, 2023	10:15	0.05	0.05	Rob	2900	8.50	13.00	17.68	21.57	20.79	0.00	0.00	0.05	0.00	0.00	0.00	0.05	0.00	
Friday, September 1, 2023	10:18	0.38	0.38	Rob	2900	9.00	13.00	70.00	32.65	35.52	0.00	0.00	0.33	0.00	0.00	0.00	0.33	0.00	
Friday, September 1, 2023	11:17	1.37	1.37	Rob	2900	7.99	12.84	70.05	44.64	48.16	0.00	9.50	0.98	9.66	0.00	0.00	0.98	0.00	
Friday, September 1, 2023	12:25	2.50	2.50	Rob	2900	7.15	12.84	69.94	44.66	47.95	0.00	15.00	1.13	13.24	0.00	0.00	1.13	0.00	
Friday, September 1, 2023	14:02	4.12	4.12	Rob	2900	7.89	12.84	69.94	45.52	49.07	0.00	26.00	1.62	16.08	0.00	27.00	1.62	16.70	
Friday, September 1, 2023	15:51	5.93	5.93	Rob	2900	7.73	12.78	69.92	44.76	48.17	0.00	37.00	1.82	20.37	0.00	44.00	1.82	24.22	
Monday, September 4, 2023	7:42	69.78	69.78	Rob	2900	8.97	12.50	70.05	47.79	51.79	0.00	1270.00	63.85	19.89	0.00	1275.00	63.85	19.97	
Monday, September 4, 2023	11:11	73.27	73.27	Rob	2900	9.12	12.50	70.04	47.06	50.99	0.00	60.00	3.48	17.22	0.00	44.00	3.48	12.63	
Monday, September 4, 2023	12:55	75.00	75.00	Rob	2900	9.04	12.50	70.14	48.79	52.82	0.00	27.00	1.73	15.58	0.00	23.00	1.73	13.27	
Monday, September 4, 2023	14:10	76.25	76.25	Rob	2900	9.19	12.53	70.34	48.42	52.47	0.00	21.00	1.25	16.80	0.00	21.00	1.25	16.80	
Monday, September 4, 2023	15:41	77.77	77.77	Rob	2900	9.19	12.49	70.22	49.32	53.41	0.00	24.00	1.52	15.82	0.00	21.00	1.52	13.85	
Tuesday, September 5, 2023	7:41	93.77	93.77	Rob	2900	6.74	12.50	70.18	43.98	47.27	0.00	242.00	16.00	15.12	0.00	240.00	16.00	15.00	
Tuesday, September 5, 2023	9:20	95.42	95.42	Rob	2900	7.10	12.50	70.06	43.65	46.96	0.00	22.50	1.65	13.64	0.00	22.00	1.65	13.33	
Tuesday, September 5, 2023	10:24	96.48	96.48	Rob	2900	6.67	12.50	69.94	43.78	46.98	0.00	13.50	1.07	12.66	0.00	10.50	1.07	9.84	
Tuesday, September 5, 2023	12:22	98.45	98.45	Rob	2900	9.76	12.50	70.32	48.36	52.53	0.00	37.00	1.97	18.81	0.00	33.00	1.97	16.78	
Tuesday, September 5, 2023	13:42	99.78	99.78	Rob	2900	6.68	12.50	69.95	43.62	46.87	0.00	15.00	1.33	11.25	0.00	9.50	1.33	7.13	
Tuesday, September 5, 2023	14:37	100.70	100.70	Rob	2900	9.57	12.42	70.28	50.08	54.34	0.00	13.50	0.92	14.73	0.00	12.50	0.92	13.64	
Tuesday, September 5, 2023	15:45	101.83	101.83	Rob	2900	6.95	12.49	70.24	43.27	46.50	0.00	18.50	1.13	16.32	0.00	21.50	1.13	18.97	
Wednesday, September 6, 2023	7:42	117.78	117.78	Rob	2900	9.75	12.52	70.31	49.37	53.63	0.00	223.00	15.95	13.98	0.00	214.00	15.95	13.42	
Wednesday, September 6, 2023	9:35	119.67	119.67	Rob	2900	6.73	12.49	70.19	46.01	49.40	0.00	23.00	1.88	12.21	0.00	16.50	1.88	8.76	
Wednesday, September 6, 2023	10:48	120.88	120.88	Rob	2900	9.65	12.50	70.21	50.25	54.51	0.00	21.00	1.22	17.26	0.00	24.00	1.22	19.73	
Wednesday, September 6, 2023	11:50	121.92	121.92	Rob	2900	9.84	12.50	70.15	47.08	51.21	0.00	17.50	1.03	16.94	0.00	19.50	1.03	18.87	
Wednesday, September 6, 2023	13:01	123.10	123.10	Rob	2900	9.57	12.48	69.98	46.64	50.70	0.00	17.00	1.18	14.37	0.00	13.00	1.18	10.99	
Wednesday, September 6, 2023	14:15	124.33	124.33	Rob	2900	9.77	12.49	70.23	49.56	53.77	0.00	19.00	1.23	15.41	0.00	17.00	1.23	13.78	
Wednesday, September 6, 2023	15:50	125.92	125.92	Rob	2900	9.68	12.49	70.39	50.60	54.89	0.00	21.00	1.58	13.26	0.00	14.00	1.58	8.84	
Thursday, September 7, 2023	7:48	141.88	141.88	Rob	2900	9.79	12.49	70.07	48.83	53.02	0.00	244.00	15.97	15.28	0.00	222.00	15.97	13.90	
Thursday, September 7, 2023	10:15	144.33	144.33	Rob	2900	9.73	12.54	70.49	49.49	53.73	0.00	40.00	2.45	16.33	0.00	34.50	2.45	14.08	
Thursday, September 7, 2023	11:13	145.30	145.30	Rob	0	0.00													
Thursday, September 7, 2023	11:14	145.32	145.32	Rob															

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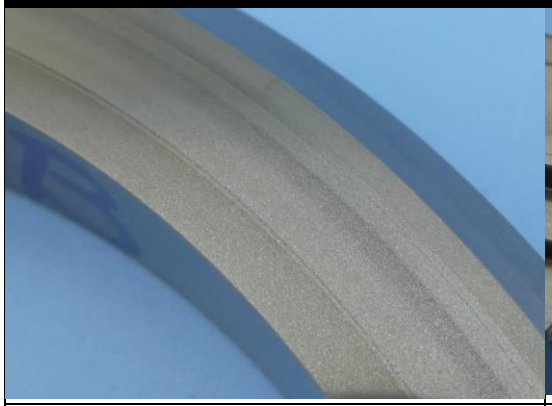
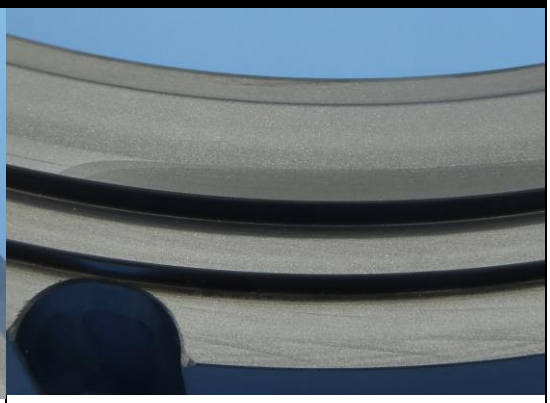
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Measurements Results

Leakage rate		
	Inner Seal	Outer Seal
Total Run time [h]	145.3	145.3
Total Seal Leakage [ml]	2378	2478
Seal Leakage Rate [ml/h]	16.5	17

Post-Test Pictures

IB Rotor	IB Stator
	
Ra	Ra

OB Rotor	OB Stator
	
Ra	Ra

Test Conclusion

The second test results showed very stable seal performance using Ethylene Glycol and water mixture, the leakage was low and within acceptable limits.

Seal faces didn't show any damage or deterioration for the duration of the test exceeding 144h. This conclude that seal faces are suitable for Ethylene glycol and 70barg.

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